

# MARYAM ALIAKBARPOUR

## Curriculum Vitae

<https://maryamaliakbarpour.com>

maryama@rice.edu

RESEARCH     ◇ Theoretical Computer Science

INTERESTS   ◇ Statistical Learning Theory

◇ Differential Privacy

EDUCATION   ◇ **Massachusetts Institute of Technology (MIT)**

Cambridge, MA, USA

**Ph.D.** in Computer Science

Sep 2015 - Sep 2020

Thesis: Distribution Testing: Classical and New Paradigms

Advisor: **Prof. Ronitt Rubinfeld**

◇ **Massachusetts Institute of Technology (MIT)**

Cambridge, MA, USA

**M.S.** in Electrical Engineering and Computer Science

Sep 2013 - Sep 2015

Thesis: Learning and Testing Junta Distributions over Hypercubes

Advisor: **Prof. Ronitt Rubinfeld**

◇ **Sharif University of Technology**

Tehran, Iran

**B.S.** in Computer Engineering - Software

Sep 2009 - Jun 2013

WORK     ◇ Michael B. Yuen and Sandra A. Tsai **Assistant Professor**

Jul 2023 - present

EXPERIENCES   Department of Computer Science,

Department of Electrical and Computer Engineering (courtesy appointment),

Ken Kennedy Institute: AI, Data, and Computing for Global Impact,

**Rice University**

◇ **Visitor** and **Research Fellow** at **Simons Institute, UC Berkeley**

Long-Term Participant of Federated and Collaborative Learning Program

Spring 2026

Research fellow of *Sublinear Algorithms Program*

Summer 2024

Visiting participant of *Probability, Geometry, and Computation in High Dim. Program*

Fall 2020

◇ **Postdoctoral** Scholar at **Boston University/Northeastern University**

Sep 2021 - Jun 2023

◇ **Postdoctoral** Research Associate at **UMass Amherst**

Sep 2020 - Aug 2021

◇ Summer internship at **Google**

Summer 2017

Sunnyvale, CA, USA

◇ Summer internship at **EPFL** (Ecole Polytechnique Federale de Lausanne),

Summer 2012

Lausanne, Switzerland.

RESEARCH   1. *High-Dimensional Robust Mean Estimation with Untrusted Batches*

PUBLICATIONS   **M. Aliakbarpour**, V. Braverman, Y. Liu, J. Yin

Submitted.

2. *Shadow Tomography Against Adversaries*

**M. Aliakbarpour**, V. Braverman, N. Chia, C. Lin, Y. Liu, A. Oufkir, Y. Shen

Submitted.

3. *Auditing Information Disclosure During LLM-Scale Gradient Descent Using Gradient Uniqueness*

- S. Abdelghafar, **M. Aliakbarpour**, C. Jermaine  
Submitted.
4. *How fast can you find a good hypothesis?*  
A. Aamand, **M. Aliakbarpour**, J. Y. Chen, S. Silwal,  
To appear in 39-th Annual Conference on Learning Theory, **COLT 2026**
  5. *Optimal Prediction-Augmented Algorithms for Testing Independence of Distributions*  
**M. Aliakbarpour**, A. Azizi, R. Steven,  
To appear in 39-th Annual Conference on Learning Theory, **COLT 2026**
  6. *High-Probability Bounds For Heterogeneous Local Differential Privacy*  
**M. Aliakbarpour**, A. Fallah, S. Roy, R. Stevens  
29th International Conference on Artificial Intelligence and Statistics, **AISTATS 2026**.
  7. *Support Basis: Fast Attention Beyond Bounded Entries*  
**M. Aliakbarpour**, V. Braverman, J. Yin, H. Zhang  
29th International Conference on Artificial Intelligence and Statistics, **AISTATS 2026**.  
Selected for **Spotlight** Presentation; 2.5% acceptance rate
  8. *On the Structure of Replicable Hypothesis Testers*  
A. Aamand, **M. Aliakbarpour**, J. Y. Chen, S. Silwal, S. Narayanan  
37th ACM-SIAM Symposium on Discrete Algorithms, **SODA 2026**
  9. *Nearly-Linear Time Private Hypothesis Selection with the Optimal Approximation Factor*  
**M. Aliakbarpour**, Z. Shi, R. Stevens, V. Wang  
39th Conference on Neural Information Processing Systems, **NeurIPS 2025**
  10. *Adversarially Robust Quantum State Learning and Testing*  
**M. Aliakbarpour**, V. Braverman, N. Chia, Y. Liu  
66th IEEE Symposium on Foundations of Computer Science, **FOCS 2025**
  11. *Better Private Distribution Testing by Leveraging Unverified Auxiliary Data*  
**M. Aliakbarpour**, A. Burudgunte, C. Cannone, R. Rubinfeld  
38th Annual Conference on Learning Theory **COLT 2025**  
Presented in Theory and Practice of Differential Privacy, **TPDP 2025**
  12. *Enhancing Feature-Specific Data Protection via Bayesian Coordinate Differential Privacy*  
**M. Aliakbarpour** S. Chaudhuri, T. A. Courtade, A. Fallah, M. I. Jordan  
28th International Conference on Artificial Intelligence and Statistics, **AISTATS 2025**
  13. *Privacy in Metalearning and Multitask Learning: Modeling and Separations*  
**M. Aliakbarpour**, K. Bairaktari, A. Smith, M. Swanberg, J. Ullman  
28th International Conference on Artificial Intelligence and Statistics, **AISTATS 2025**
  14. *Optimal Algorithms for Augmented Testing of Discrete Distributions*  
**M. Aliakbarpour**, P. Indyk, R. Rubinfeld, S. Silwal  
38th Conference on Neural Information Processing Systems, **NeurIPS 2024**
  15. *Optimal Hypothesis Selection in (Almost) Linear Time*  
**M. Aliakbarpour**, M. Bun, A. Smith  
38th Conference on Neural Information Processing Systems, **NeurIPS 2024**
  16. *Metalearning with Very Few Samples Per Task*  
**M. Aliakbarpour**, K. Bairaktari, G. Brown, A. Smith, J. Ullman  
37th Annual Conference on Learning Theory, **COLT 2024**
  17. *Differentially Private Medians and Interior Points for Non-Pathological Data*  
**M. Aliakbarpour**, R. Silver, T. Steinke, J. Ullman  
15th Innovations in Theoretical Computer Science **ITCS 2024**  
Presented in Theory and Practice of Differential Privacy, **TPDP 2023**
  18. *Hypothesis Selection with Memory Constraints*  
**M. Aliakbarpour**, M. Bun, A. Smith  
37th Conference on Neural Information Processing Systems, **NeurIPS 2023**

19. *Testing Tail Weight of a Distribution Via Hazard Rate*  
**M. Aliakbarpour**, A.S. Biswas, K. Ravichandran, R. Rubinfeld  
 34th International Conference on Algorithmic Learning Theory, **ALT 2023**
20. *Estimation of Entropy in Constant Space with Improved Sample Complexity*  
**M Aliakbarpour**, A. McGregor, J. Nelson, E. Waingarten  
 36th Conference on Neural Information Processing Systems, **NeurIPS 2022**
21. *Local Differential Privacy Is Equivalent to Contraction of an  $f$ -Divergence*  
 S. Asodeh, **M. Aliakbarpour**, F. Calmon  
 2021 IEEE International Symposium on Information Theory, **ISIT 2021**
22. *Rapid Approximate Aggregation with Distribution-Sensitive Interval Guarantees*  
 S. Macke, **M. Aliakbarpour**, I. Diakonikolas, A. Parameswaran, R. Rubinfeld  
 37th IEEE International Conference on Data Engineering, **ICDE 2021**
23. *Testing Determinantal Point Processes*  
 Khashayar Gatmiry, **M. Aliakbarpour**, Stefanie Jegelka  
 34th Conference on Neural Information Processing Systems, **NeurIPS 2020**  
 Selected for **Spotlight** Presentation
24. *Testing Properties of Multiple Distributions with Few Samples*  
**M. Aliakbarpour**, S. Silwal  
 11th Innovations in Theoretical Computer Science Conference, **ITCS 2020**
25. *Private Testing of Distributions via Sample Permutations*  
**M. Aliakbarpour**, I. Diakonikolas, D. Kane, R. Rubinfeld  
 33rd Conference on Neural Information Processing Systems, **NeurIPS 2019**
26. *Towards Testing Monotonicity of Distributions Over General Posets*  
**M. Aliakbarpour**, T. Gouleakis, J. Peebles, R. Rubinfeld, A. Yodpinyanee  
 32nd Annual Conference on Learning Theory, **COLT 2019**
27. *Testing Mixtures of Distributions*  
**M. Aliakbarpour**, R. Kumar, R. Rubinfeld  
 32nd Annual Conference on Learning Theory, **COLT 2019**
28. *Differentially Private Identity and Equivalence Testing of Discrete Distributions*  
**M. Aliakbarpour**, I. Diakonikolas, R. Rubinfeld  
 35th International Conference on Machine Learning, **ICML 2018**, pp. 169–178
29. *Sublinear-Time Algorithms for Counting Star Subgraphs via Edge Sampling*  
**M. Aliakbarpour**, A. S. Biswas, T. Gouleakis, J. Peebles, R. Rubinfeld, A. Yodpinyanee  
**Algorithmica 2018**, pp. 668–697
30. *I’ve Seen “Enough”: Incrementally Improving Visualizations to Support Rapid Decision Making*  
 S. Rahman, **M. Aliakbarpour**, H. Kong, E. Blais, K. Karahalios, A. G. Parameswaran, R. Rubinfeld  
 43rd International Conference on Very Large Data Bases, **VLDB 2017**, pp. 1262–1273
31. *Learning and Testing Junta Distributions*  
**M. Aliakbarpour**, E. Blais, R. Rubinfeld  
 29th Annual Conference on Learning Theory, **COLT 2016**, pp. 19–46
32. *Join of Two Graphs has a Nowhere-zero 3-flow*  
 S. Akbari, **M. Aliakbarpour**, N. Ghanbari, E. Nategh, H. Shahmohamad  
**Czechoslovak Mathematical Journal 2014**, pp. 433–446
33. *Minimum flow number of complete multipartite graphs*  
 S. Akbari, **M. Aliakbarpour**, N. Ghanbari, E. Nategh, H. Shahmohamad  
**Bulletin of the Institute of Combinatorics and its Applications 2012**, pp. 57–64

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|-------------------|----------------------------------------------------------------|------|
| HONORS AND AWARDS | ◇ Career Champion Award for the Class of 2024, Rice University | 2024 |
|                   | ◇ Selected participant of <b>Rising Stars in EECS</b>          | 2018 |

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|------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                              | ◇ <b>Neekeyfar Award</b> , Office of Graduate Education, MIT                                                    | 2013                                  |
|                              | ◇ <b>Ranked 2<sup>nd</sup></b> in Cumulative GPA                                                                | 2013                                  |
|                              | among the students in Computer Engineering Department who started in Fall 2009, Sharif University of Technology |                                       |
|                              | ◇ <b>Ranked 9<sup>th</sup></b> in Nationwide Graduate Entrance Qualification Exam                               | 2012                                  |
|                              | computer engineering (artificial intelligence discipline)                                                       |                                       |
|                              | among more than 31,000 participants, Iran                                                                       |                                       |
|                              | ◇ <b>Silver Medal</b> in Iranian National Olympiad in Informatics                                               | 2008                                  |
| INVITED<br>TALKS &<br>PANELS | ◇ Workshop on Trust in Decentralized Systems, Simons Institute, UC Berkeley                                     | Mar 2026                              |
|                              | ◇ Workshop on Learning from Heterogeneous Sources, Simons Institute, UC Berkeley                                | Feb 2026                              |
|                              | ◇ Probability Seminar, University of California, Davis                                                          | Jan 2026                              |
|                              | ◇ Workshop on Local Algorithms (WoLA)                                                                           | Aug 2025                              |
|                              | ◇ Workshop on Algorithms for Large Data WALDO                                                                   | Apr 2025                              |
|                              | ◇ Panelist for the Center for Engineering Excellence Through Equity, Rice University                            | Jan 2025                              |
|                              | ◇ Invited mentor at Mentoring Tables, Learning Theory Alliance                                                  | Dec 2024                              |
|                              | ◇ Panelist at Rising Stars Workshop, MIT                                                                        | Oct 2024                              |
|                              | ◇ MIT, Algorithms & Complexity Seminar                                                                          | Oct 2024                              |
|                              | ◇ Rice University, Department of Statistics, STAT Colloquia                                                     | Sep 2024                              |
|                              | ◇ University of Texas at Austin, Computer Science Department, Theory Seminar Series                             | Sep 2024                              |
|                              | ◇ Workshop on Extroverted Sublinear Algorithms, Simons Institute, UC Berkeley                                   | Jun 2024                              |
|                              | ◇ Workshop on Local Algorithms (WOLA), MIT                                                                      | Aug 2023                              |
|                              | ◇ MIT, Algorithms & Complexity Seminar                                                                          | Apr 2023                              |
|                              | ◇ Purdue University, Theory seminar                                                                             | Nov 2022                              |
|                              | ◇ Sublinear Algorithm Workshop, FODSI, MIT                                                                      | Aug 2022                              |
|                              | ◇ Workshop on Differential Privacy and Statistical Data Analysis, Fields Institute                              | Jul 2022                              |
|                              | ◇ Workshop on Local Algorithms (WOLA)                                                                           | Jun 2022                              |
|                              | ◇ Northeastern University                                                                                       | Nov 2021                              |
|                              | ◇ Boston University                                                                                             | Nov 2021                              |
|                              | ◇ Learning and Testing in High Dimensions Workshop, Simons Institute, Berkeley                                  | Dec 2020                              |
|                              | ◇ Carnegie Mellon University, Theory lunch                                                                      | Oct 2020                              |
|                              | ◇ Harvard University, DP meeting                                                                                | Sep 2020                              |
|                              | ◇ Workshop on Local Algorithms (WOLA)                                                                           | Jul 2020                              |
|                              | ◇ Georgia Tech, ARC Colloquium                                                                                  | Mar 2020                              |
|                              | ◇ University of Massachusetts Amherst                                                                           | Feb 2020                              |
|                              | ◇ Boston University                                                                                             | Feb 2020                              |
|                              | ◇ Northeastern University                                                                                       | Oct 2018                              |
|                              | ◇ IBM Thomas J. Watson Research Center                                                                          | Dec 2016                              |
| TEACHING<br>EXPERIENCES      | ◇ <b>Instructor</b> , Rice University:                                                                          |                                       |
|                              | · Reasoning About Algorithms                                                                                    | Fall 2025                             |
|                              | · Graduate Seminar in Learning Theory                                                                           | Fall 2023, Fall 2024                  |
|                              | · Probabilistic Toolkit for Learning and Computing                                                              | Spring 2024, Spring 2025, Spring 2026 |
|                              | ◇ <b>Teaching Assistant</b> , Massachusetts Institute of Technology:                                            |                                       |
|                              | · Geometric Computation                                                                                         | Spring 2020                           |
|                              | · Introduction to Algorithms                                                                                    | Fall 2017                             |
|                              | · Design and Analysis of Algorithms                                                                             | Spring 2016, Fall 2016                |
|                              | ◇ <b>Teaching Assistant</b> , Sharif University of Technology:                                                  |                                       |
|                              | · For six times in Algorithms, Discrete Mathematics, Scientific and Technical Presentation.                     |                                       |

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| EXTERNAL<br>SERVICE                   | <ul style="list-style-type: none"> <li>◇ <b>Program committee / Area chair:</b> FOCS 2026, RANDOM 2026, STOC 2026, AISTAT 2026 , TPDP 2025, AISTAT 2025, FOCS 2024, COLT 2024, ITCS 2024, TPDP 2023, COLT 2021, ITCS 2022, COLT 2022</li> <li>◇ Reviewer committee: COLT 2020, NeurIPS 2020, ICLR 2021, FAccT 2022</li> <li>◇ Reviewer and sub-reviewer for various conferences and journals in theoretical computer science and machine learning</li> <li>◇ NSF proposal panelist</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| LEADERSHIP<br>ROLES AND<br>ACTIVITIES | <ul style="list-style-type: none"> <li>◇ <b>Leader of “TRUST-AI: Foundations of Trustworthy AI” Research Cluster</b> Jun 2025–<br/>Co-founded and led a research cluster focused on privacy, fairness, security, and societal impact of AI at Ken Kennedy Institute. This cluster led to the development of two NSF research proposals and collaborations directly aligned with cluster themes.</li> <li>◇ <b>Co-organizer of Crossroads of AI and Society: Incentives, Privacy, &amp; Fairness</b> Jun 2026<br/>This forthcoming workshop will be held at the Rice Global Paris Center and will focus on the broad theme of AI alignment. Funding for the workshop has been secured from Rice Global Paris, the Creative Ventures Fund, and the Ken Kennedy Institute TRUST-AI Cluster.</li> <li>◇ <b>Co-organizer of Texas Area Theory and Algorithms Day</b> Sep 2025<br/>A one-day workshop that aims to strengthen connection between researchers in theoretical computer science (TCS) and algorithms across Texas and the surrounding regions. The workshop brought together researchers from different institutions to share ideas, establish collaborations, and provide exposure to junior researchers. For more info, visit our website.</li> <li>◇ <b>Co-organizer of Applied Algorithms for Machine Learning Workshop</b> Jun 2024<br/>A workshop in Rice Global Paris Center featuring an exceptional lineup of speakers and attracting over 80 participants to explore cutting-edge topics at the intersection of algorithms and machine learning. For more info, visit our website.</li> <li>◇ <b>Co-organizer of Boston-Area Data Privacy Seminar</b> Sep 2021 – Dec 2022<br/>Jointly organized by Boston University, Northeastern University, and Harvard University, this seminar series featured speakers from diverse backgrounds discussing recent and impactful research on the foundations of data privacy and related topics.</li> <li>◇ <b>Member of Resources for Easing Friction and Stress (REFS)</b> 2016 - 2019<br/>Department of Electrical Engineering and Computer Science, MIT, Cambridge, MA, USA<br/>REFS is a group of EECS graduate students trained as peer mediators by Conflict Management at MIT. Our role is to support the graduate community and serve as the first point of contact in dealing with stress and conflict.</li> <li>◇ <b>Member of Sidney Pacific Executive Council (SPEC)</b> 2015 – 2016<br/>Sidney Pacific Graduate Community, MIT, Cambridge, MA, USA<br/>Elected as <i>Chair of the Halls</i> in Sidney Pacific, my graduate dormitory with over 600 residents, focusing on the health and wellness of students. I trained and led a group of graduate students, the hall councilors. Our team’s goal was to build smaller communities within the dorm and ensure that every resident had someone to reach out to. Additionally, I organized several health and wellness events for the residents.</li> </ul> |